Worksheet: Strategic Price Setting

ECON 201

Definitions

- **Best response**: the strategy that gives a player the highest pay-off, given what the others choose.
- **Nash equilibrium**: a set of strategies, one for each player, where each player's strategy is a best response to the others'.
- **Dominant strategy**: a strategy that is a player's best response whatever the others choose.
- **Prisoners' dilemma**: a game where each player has a dominant strategy, but another outcome gives every player a higher pay-off.

Wanda and Kit, with fewer loyal customers

Each rental costs \$10, and each firm picks $H = \$36$ or $L = \$20$. Now each firm has **7 loyal** customers, **13 switchers**, and **10 bargain hunters**. When one firm is cheaper, the other firm's switchers and bargain hunters come to it.

1. Fill in the last two columns of Wanda's customers and profit. The first two are done for you.

Wanda's price, Kit's price	H, H	L, L	L, H	H, L
Her loyal customers	7	7		
Her switchers	13	13		
Kit's switchers	0	0		
Her bargain hunters	0	10		
Kit's bargain hunters	0	0		
Total customers	20	30		
Profit, $(P - 10)\times$ total	520	300		

2. Kit's numbers mirror Wanda's. Fill in the pay-off matrix, with Wanda's profit first in each cell.

	Kit picks H	**Kit picks** L
Wanda picks H	520, 520	
Wanda picks L		300, 300

3. If Kit picks H, Wanda earns _____ at H and _____ at L, so her best response is _____.

If Kit picks L, Wanda earns _____ at H and _____ at L, so her best response is _____.

4. Does Wanda have a dominant strategy? _____ The Nash equilibrium is _____, with profits of _____ each.

5. Is there another outcome where *both* firms would earn more? Which one? What kind of game is this?

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